

Exercise Sheet 3: Sketch Solutions

Advanced Microeconomics: Economic Decision Making

Question 1

Let u be three times continuously differentiable with $u' > 0$, and let ε be a zero-mean risk with variance σ^2 and small support around 0; initial wealth x . Define the risk premium ρ by $u(x - \rho) = \mathbb{E}[u(x + \varepsilon)]$.

- (a) Expand both sides to second order in small quantities and derive $\rho \approx \frac{1}{2}r_A(x)\sigma^2$ where $r_A = -u''/u'$. [15]
- (b) For proportional risks $\varepsilon = x\tilde{\varepsilon}$ with $\tilde{\varepsilon}$ zero-mean of variance $\tilde{\sigma}^2$, deduce $\rho/x \approx \frac{1}{2}r_R(x)\tilde{\sigma}^2$. [5]
- (c) In one sentence each: why is the premium quadratic in the size of the risk, and what did this quadratic imply for the course's ex-ante cost-of-inflation numbers? [5]

Solution. (a) Right side: $\mathbb{E}[u(x+\varepsilon)] = u(x) + u'(x)\mathbb{E}[\varepsilon] + \frac{1}{2}u''(x)\mathbb{E}[\varepsilon^2] + o(\sigma^2) = u(x) + \frac{1}{2}u''(x)\sigma^2 + o(\sigma^2)$.

Left side: $u(x - \rho) = u(x) - u'(x)\rho + o(\rho)$. Equate and solve: $\rho = -\frac{1}{2}\frac{u''(x)}{u'(x)}\sigma^2 + o(\sigma^2) = \frac{1}{2}r_A(x)\sigma^2$.

(b) $\text{Var}(x\tilde{\varepsilon}) = x^2\tilde{\sigma}^2$, so $\rho \approx \frac{1}{2}r_A(x)x^2\tilde{\sigma}^2$; divide by x : $\rho/x = \frac{1}{2}[xr_A(x)]\tilde{\sigma}^2 = \frac{1}{2}r_R(x)\tilde{\sigma}^2$.

(c) The first-order term vanishes because the risk is zero-mean — only variance survives, and variance is quadratic. Consequently the pure price of *not knowing* is second-order small relative to typical realized losses, which is exactly why the course's ex-ante premia (0.1–0.2% of expenditure) sat an order below the realized bills.

Question 2

A worker's real income lottery is: uninsured, -1% w.p. 0.75 and -9% w.p. 0.25; with an indexation contract, $+0.5\%$ w.p. 0.75 and -1.5% w.p. 0.25. Separately, the contract raises the probability of a job loss worth -40% of income by 1pp in the bad state; treat this as an expected cost computed *outside* the lottery. CRRA, $r = 2$.

- (a) Compute both certainty equivalents and the gross insurance value ΔCE . [12]
- (b) Decompose ΔCE into a mean part and a risk part, and interpret the split. [8]
- (c) Net off the job-cost term and state the final valuation with its three components. [5]

Solution. (a) Uninsured (gross incomes 0.99, 0.91): mean 0.97; $\mathbb{E}[u] = -(0.75/0.99 + 0.25/0.91) = -1.032301$; $c_U = 0.968709$. Indexed (1.005, 0.985): mean 1.0; $\mathbb{E}[u] = -(0.746269 + 0.253807) = -1.000076$; $c_I = 0.999924$. $\Delta CE = +3.12\%$.

(b) $\Delta \text{mean} = 1.0 - 0.97 = +3.00\%$; risk part = $\Delta CE - \Delta \text{mean} = +0.12\%$ (equivalently $\rho_U - \rho_I =$

0.129% – 0.008%). The contract’s value is overwhelmingly a *transfer arriving in the right state*; the variance-removal service the word “insurance” suggests is worth 0.12pp — the $\frac{1}{2}r\sigma^2$ logic of P5 made concrete.

(c) Job cost = $0.25 \times 1\text{pp} \times 40\% = 0.10\%$ expected. Net value = $3.12 - 0.10 = +3.0\%$, composed of +3.0 mean transfer + 0.12 risk removal – 0.10 employment-hazard cost: the valuation’s sign is carried by the transfer, so its assessment must confront who finances the transfer — incidence, not variance mathematics.

Question 3

$u(x) = -e^{-ax}$ with $a = 0.01$; initial wealth w ; the agent faces a fair coin paying ± 100 .

- (a) Derive the exact certainty-equivalent formula $c = w - \frac{1}{a} \ln\left(\frac{1}{2}e^{a \cdot 100} + \frac{1}{2}e^{-a \cdot 100}\right)$ and evaluate the risk premium. [12]
- (b) Compare with $\frac{1}{2}a\sigma^2$ and explain the direction of the discrepancy. [7]
- (c) Verify explicitly that the premium does not depend on w , and state which behavioural implication of CARA this illustrates (and why it is empirically awkward). [6]

Solution. (a) $\mathbb{E}[u] = -\frac{1}{2}(e^{-a(w+100)} + e^{-a(w-100)}) = -e^{-aw} \cosh(a \cdot 100)$. Set $-e^{-ac} = \mathbb{E}[u]$: $c = w - \frac{1}{a} \ln \cosh(1) = w - 100 \times 0.4338 = w - 43.4$. Premium = $\mathbb{E}[x] - c = w - (w - 43.4) = 43.4$.

(b) $\frac{1}{2}a\sigma^2 = \frac{1}{2}(0.01)(10000) = 50$: the quadratic approximation *overshoots* the exact 43.4 here; with $a\varepsilon = 1$ the risk is far outside the local regime and higher-order (even-moment) corrections, which for CARA’s exponential pull the exact premium below the quadratic, are material.

(c) w cancelled identically in (a): the premium for an absolute gamble is the same for a student and a billionaire. That is CARA’s signature — and its awkwardness: measured behaviour shows richer agents treating a fixed-euro gamble as smaller, which is why the course’s proportional (CRRA) family carried the applications.